Frozen encoders alone: WavLM-large, wav2vec2-large-lv60, XLS-R 300M

FINAL_V1 experiment group summary

Protocol FINAL_V1 Status: **COMPLETED** Generated: 2026-10-20T11:51:04Z

Results

- 12 of 12 conditions in this group are complete.
- Highest pooled AUPRC (descriptive, selection-biased among 12 conditions): XLSR Logistic regression, AUPRC 1.000, AUROC 1.000.
- Lowest: WLM RBF SVM, AUPRC 0.887, AUROC 0.873.
- Range of AUROC over the group: 0.873 to 1.000; median 0.995.
- Speaker errors at threshold 0.5 (false negatives + false positives) per condition: minimum 1, median 1, maximum 6 (out of 47 speakers).
- Fold variability: median over conditions of the mean fold AUROC is 1.000 with a median SD of 0.000 over the five outer folds (9 to 10 speakers per fold).

Reading the table

- Primary metrics are pooled speaker-level out-of-fold AUPRC and AUROC (47 speakers: 25 PD, 22 controls; prevalence 53.2% PD); F1, sensitivity, specificity and balanced accuracy use the fixed threshold 0.5.
- Intervals are percentile bootstrap intervals over speakers (2,000 resamples); with 47 speakers one person moves a metric by roughly 2 points. Comparisons are descriptive; no significance is claimed.
- Recording date and class are confounded in this corpus (every date holds one class; all 44.1 kHz audio is PD), so a high score does not show disease detection.

Table 1: Pooled out-of-fold speaker-level results

Representation	Model	AUPRC [95% CI]	AUROC [95% CI]	F1	Sens.	Spec.	Bal.acc.
WLM	Logistic regression	0.996 [0.981, 1.000]	0.995 [0.977, 1.000]	0.980	0.960	1.000	0.980
WLM	Linear SVM	0.996 [0.982, 1.000]	0.995 [0.977, 1.000]	0.980	0.960	1.000	0.980
WLM	RBF SVM	0.887 [0.765, 0.972]	0.873 [0.752, 0.966]	0.889	0.960	0.773	0.866
WLM	XGBoost	0.998 [0.991, 1.000]	0.998 [0.989, 1.000]	0.960	0.960	0.955	0.957
W2V	Logistic regression	0.992 [0.971, 1.000]	0.989 [0.960, 1.000]	0.980	0.960	1.000	0.980
W2V	Linear SVM	0.986 [0.948, 1.000]	0.975 [0.913, 1.000]	0.980	0.960	1.000	0.980
W2V	RBF SVM	0.997 [0.987, 1.000]	0.996 [0.984, 1.000]	0.962	1.000	0.909	0.955
W2V	XGBoost	0.989 [0.957, 1.000]	0.982 [0.935, 1.000]	0.980	0.960	1.000	0.980
XLSR	Logistic regression	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.980	0.960	1.000	0.980
XLSR	Linear SVM	0.997 [0.987, 1.000]	0.996 [0.984, 1.000]	0.980	0.960	1.000	0.980
XLSR	RBF SVM	0.986 [0.950, 1.000]	0.976 [0.919, 1.000]	0.941	0.960	0.909	0.935
XLSR	XGBoost	0.997 [0.986, 1.000]	0.996 [0.983, 1.000]	0.980	0.960	1.000	0.980